



Designation: E283/E283M – 19

Standard Test Method for Determining Rate of Air Leakage Through Exterior Windows, Skylights, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen¹

This standard is issued under the fixed designation E283/E283M; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ϵ) indicates an editorial change since the last revision or reapproval.

This standard has been approved for use by agencies of the U.S. Department of Defense.

1. Scope

1.1 This test method covers a standard laboratory procedure for determining the air leakage rates of exterior windows, skylights, curtain walls, and doors under specified differential pressure conditions across the specimen. The test method described is for tests with constant temperature and humidity across the specimen. Persons interested in performing air leakage tests on units exposed to various temperature differences across the specimen should reference Test Method [E1424](#).

1.2 This laboratory procedure is applicable to exterior windows, skylights, curtain walls, and doors and is intended to measure only such leakage associated with the assembly and not the installation. The test method can be adapted for the latter purpose.

NOTE 1—Performing tests under uncontrolled conditions or with a temperature differential across the specimen may affect the air leakage rate. This is not addressed by this test method.

1.3 This test method is intended for laboratory use. Persons interested in performing field air leakage tests on installed units should reference Test Method [E783](#).

1.4 Persons interested in evaluating air permeance of building materials should reference Test Method [E2178](#).

1.5 Persons interested in determining air leakage of air barrier assemblies should reference Test Method [E2357](#).

1.6 Persons using this procedure should be knowledgeable in the areas of fluid mechanics, instrumentation practices, and shall have a general understanding of fenestration products and components.

1.7 The values stated in either SI units or inch-pound units are to be regarded separately as standard. The values stated in

each system are not necessarily exact equivalents; therefore, to ensure conformance with the standard, each system shall be used independently of the other, and values from the two systems shall not be combined.

1.8 *This standard does not purport to address all of the safety problems, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.* For specific hazard statement, see Section 7.

1.9 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

2. Referenced Documents

2.1 ASTM Standards:²

[E631](#) Terminology of Building Constructions

[E783](#) Test Method for Field Measurement of Air Leakage Through Installed Exterior Windows and Doors

[E1424](#) Test Method for Determining the Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure and Temperature Differences Across the Specimen

[E2178](#) Test Method for Air Permeance of Building Materials

[E2357](#) Test Method for Determining Air Leakage Rate of Air Barrier Assemblies

2.2 ISO Standard:³

[ISO/IEC 17025](#) General requirements for the competence of testing and calibration laboratories

¹ This test method is under the jurisdiction of ASTM Committee [E06](#) on Performance of Buildings and is the direct responsibility of Subcommittee [E06.51](#) on Performance of Windows, Doors, Skylights and Curtain Walls.

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² For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

³ Available from International Organization for Standardization (ISO), ISO Central Secretariat, BIBC II, Chemin de Blandonnet 8, CP 401, 1214 Vernier, Geneva, Switzerland, <http://www.iso.org>.



3. Terminology

3.1 *Definitions*—Terms used in this test method are defined in Terminology E631.

3.2 *Definitions of Terms Specific to This Standard:*

3.2.1 *air leakage rate* (q_A or q_l), $L/(s \cdot m^2)$ [$ft^3/min \cdot ft^2$], or $L/(s \cdot m)$ [$ft^3/min \cdot ft$]*—the air leakage per unit of specimen area (A) or per unit length of operable crack perimeter (l).*

3.2.2 *extraneous air leakage* (Q_e), m^3/s [ft^3/min]*—the volume of air flowing per unit of time through the test chamber and test apparatus, exclusive of the air flowing through the test specimen, under a test pressure difference and test temperature difference, converted to standard conditions.*

3.2.2.1 *Discussion*—Extraneous leakage is the sum of all leakage other than that intended to be measured by the test.

3.2.3 *skylight*—for the purposes of this test method, skylights include roof windows, sloped glazing, tubular daylighting devices (TDDs), and units skylights.

3.2.4 *specimen*—the entire assembled unit submitted for test as described in Section 7.

3.2.5 *specimen air leakage* (Q_s), L/s [ft^3/min]*—the volume of air flowing per unit of time through the specimen under a test pressure difference and test temperature difference, converted to standard conditions.*

3.2.6 *specimen area* (A), m^2 [ft^2]*—the area determined by the overall dimensions of the frame that fits into the rough opening.*

3.2.7 *standard conditions*—in this test method, dry air at:

Pressure—101.3 kPa [29.92 in. Hg]

Temperature—20.8 °C [69.4 °F]

Air Density—1.202 kg/m³ [0.075 lbm/ft³]

3.2.8 *test pressure differences*, Pa [lbf/ft^2]*—the specified differential static air pressure across the specimen.*

3.2.9 *total air flow* (Q_t), L/s [ft^3/min]*—the volume of air flowing per unit of time through the test chamber and test apparatus, inclusive of the air flowing through the test specimen, under a test pressure difference and test temperature difference, converted to standard conditions.*

3.2.10 *unit length of operable crack perimeter* (l), m [ft]*—the sum of all perimeters of operable ventilators, sash, or doors contained in the test specimen, based on the overall dimensions of such parts. Where two such operable parts meet the two adjacent lengths of perimeter shall be counted as only one length.*

4. Summary of Test Method

4.1 The test consists of sealing a test specimen into or against one face of an air chamber, supplying air to or exhausting air from the chamber at the rate required to maintain the specified test pressure difference across the specimen, and measuring the resultant air flow through the specimen. Measurements of the resultant air flow are adjusted to standard conditions.

5. Significance and Use

5.1 This test method is a standard procedure for determining the air leakage characteristics under specified air pressure differences at ambient conditions.

NOTE 2—The air pressure differences acting across a building envelope vary greatly. The factors affecting air pressure differences and the implications or the resulting air leakage relative to the environment within buildings are discussed in the literature.⁴⁻⁶ These factors should be fully considered in specifying the test pressure differences to be used.

5.2 Rates of air leakage are sometimes used for comparison purposes. Such comparisons may not be valid unless the components being tested and compared are of essentially the same size, configuration, and design.

6. Apparatus

6.1 The description of the apparatus in this section is general in nature. Any suitable arrangement of equipment capable of maintaining the required test tolerances is permitted. (See Fig. 1.)

6.2 *Test Chamber*—A well sealed box, wall, or other apparatus into or against which the specimen is mounted and secured for testing. An air supply shall be provided to allow a positive or negative pressure differential to be applied across the specimen without significant extraneous losses. The chamber shall be capable of withstanding the differential test pressures that may be encountered in this procedure. At least one static air pressure tap shall be provided on each side of the specimen to measure the test pressure differences. The pressure tap shall be located in an area of the chamber in which pressure readings will not be affected by any supply air. The air supply opening to the chamber shall be located in an area in which it does not directly impinge upon the test specimen.

6.2.1 *Supply Air System*—A controllable blower, exhaust fan, or reversible blower designed to provide the required air flow at the specified test pressure difference. The system should provide essentially constant air flow at the specified test pressure difference for a time period sufficient to obtain readings of air flow. The inlet into the chamber (sealed box) shall be located or be shielded with a baffle so that the air flow is not directed upon the test specimen.

6.2.2 *Pressure Measuring Apparatus*—A device to measure the differential test pressures to $\pm 2\%$ of set point or ± 2.5 Pa [± 0.01 in. of water column], whichever is greater.

6.2.3 *Air Flow Metering System*—A device to measure the air flow to $\pm 5\%$ into/out of the test chamber or through the test specimen. Typically air flow is measured using a mass flow meter or volumetric flow meter. Methods to adjust volumetric flow meter measurements to standard conditions are included in Section 12.

6.2.4 *Air Temperature Measurement Apparatus*—A device to measure the temperature of the air to ± 0.5 K at the flow meter. This device may be built-in or integral to the flow meter.

6.2.5 *Absolute Pressure Measurement Apparatus*—A device to measure the absolute pressure to $\pm 2\%$ within the laboratory.

⁴ ASHRAE Handbook of Fundamentals, American Society of Heating, Refrigerating, and Air-Conditioning Engineers, Inc. (ASHRAE), Atlanta, GA, 2013.

⁵ Fluid Meters—Their Theory and Application, 5th Edition, American Society of Mechanical Engineers (ASME), New York, NY, 1959.

⁶ Power Test Code, 2nd Edition, Part 5, Chapter 4 Flow Measurements, American Society of Mechanical Engineers (ASME), New York, NY, 1956.



SKYLIGHT SPECIMEN

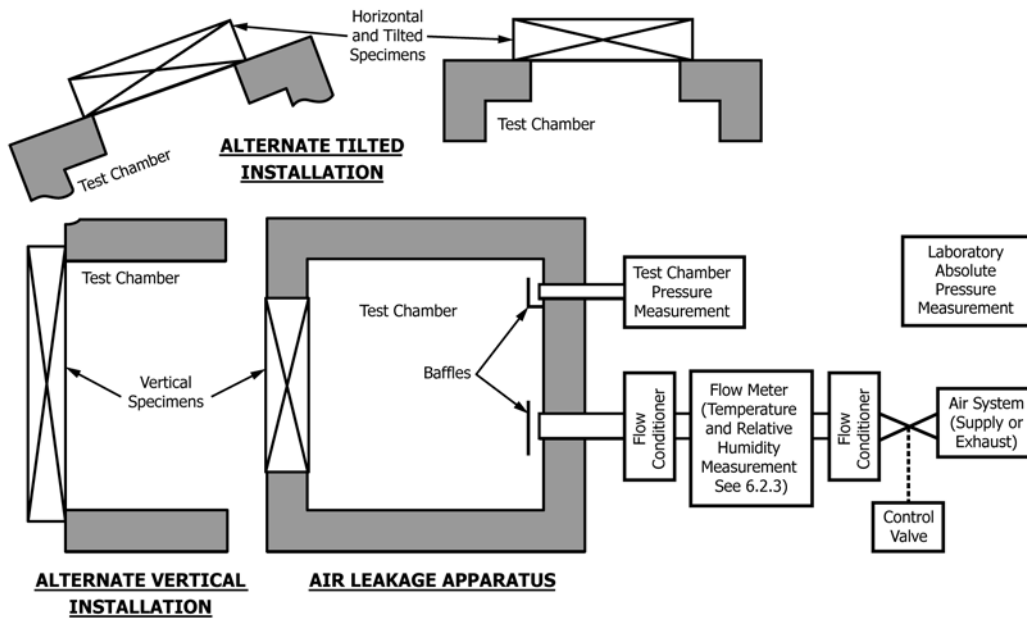


FIG. 1 General Arrangement of the Air Leakage Apparatus

6.2.6 Relative Humidity Measurement Apparatus—A device to measure the relative humidity of the air to $\pm 2\%$ RH at the flow meter or at a location where no change in the moisture content can occur between the flow meter and the relative humidity sensor. This device may be built-in or integral to the flow meter.

7. Hazards

7.1 Precaution—Glazing infill breakage may occur at the test pressure differences applied in this test. Adequate precautions should be taken to protect personnel.

8. Test Specimen

8.1 The test specimen for a wall shall be of sufficient size to determine the required performance of all typical parts of the wall system. For curtain walls, sloped glazing, or walls constructed with prefabricated units or components, the specimen width shall be not less than two typical units plus the connections and supporting elements at both sides, and sufficient to provide full loading on at least one typical vertical joint or framing member, or both.

8.1.1 For curtain wall specimens, the height shall be not less than the full building story height or the height of the unit, whichever is greater, and shall include at least one full horizontal joint, accommodating vertical expansion, such joint being at or near the bottom of the specimen, as well as all connections at top and bottom of the units. All parts of the wall test specimen shall be full size using the same materials, details, and methods of construction and anchorage as used on the actual building.

8.1.2 Conditions of structural support shall be simulated as accurately as possible.

8.2 The test specimen for a window, skylight, door, or other component shall consist of the entire assembled unit, including product frame and anchorage type and location as supplied by the manufacturer for installation in the building. If only one specimen is to be tested the selection shall be determined by the specifying authority.

9. Validation

9.1 Validation shall be performed by mounting a plywood or similar rigid panel to the test chamber in place of a test specimen, using the same mounting procedures as used for standard specimens. The panel shall be 19 ± 3 mm [$3/4 \pm 1/8$ in.] thick. The edge of the aperture supporting the orifice plate shall not be less than 50 mm [2 in.] from the nearest edge of the aperture in the orifice plate. The overall dimension of the orifice plates shall sufficiently overlap the supporting aperture to enable adequate sealing. The panel shall be attached to a minimum 140 mm [$5\frac{1}{2}$ in.] deep [nominal 2 by 6] pine test frame (buck) with dimensions 1220 mm wide by 1830 mm high [4 ft wide by 6 ft high]. The test frame and panel shall be sealed at all joints.

9.2 Each orifice plate shall be constructed of 3 mm [$1/8$ in.] thick stainless steel having an outside dimension sufficient to cover the aperture of the panel as specified in 9.1, and interior square edge diameters of the orifices are 6.35 mm [0.250 in.], 12.70 mm [0.500 in.], 25.40 mm [1.000 in.], 38.10 mm [1.500 in.], 50.80 mm [2.000 in.], and 63.50 mm [2.500 in.] as needed. The dimensions of the orifice plate shall be National Institute of Standards and Technology (NIST) traceable or verified by an ISO/IEC 17025 accredited calibration laboratory.

9.3 Fasten the orifice plate to the panel, centered over the panel aperture. Seal the hole in the orifice plate with a suitable

adhesive tape so that an extraneous reading on the air flow system can be obtained. Measure the amount of such leakage with the orifice plate sealed, at the air pressure difference as referenced in 10.1.2. After determining the extraneous leakage, remove the adhesive tape from the hole in the orifice plate and repeat the process to determine the total measured flow. The measured air flows shall be corrected to standard conditions (see Section 12.)

9.4 Validation of the air leakage test equipment shall consist of determining the flow through the air flow system to be validated using all applicable orifice plate sizes for the design range of the flow metering apparatus. The orifice plate to be used for the corresponding air flow ranges is indicated in Table 1. When needed, additional orifice plate sizes may be used to adequately cover the air flow system measuring range outside of Table 1 nominal flow ranges. It is also allowed to use other means of validation for the air leakage equipment given that it can be demonstrated that the alternative method yields accuracy of validation equal to or exceeding that of the orifice plate method. For ranges of nominal flow between 3.47 L/s [7.36 ft³/min] to 21.33 L/s [45.20 ft³/min], the use of the corresponding orifice plates referenced in Table 1 is mandatory.

NOTE 3—A minimum of three orifice plates are used to allow the air flow measuring equipment to be used for a variety of specimen sizes and chamber/wall setups.

NOTE 4—At test pressure other than 75 Pa [1.57 psf], the laboratory shall validate the airflow measuring equipment with the applicable orifice plates and record the measurements at the specified pressure(s). Using pressures greater than 75 Pa [1.57 psf] may not permit reproducibility between laboratories, nor may it warrant meeting calibration tolerance requirements as specified at 75 Pa [1.57 psf].

9.5 The air flow measuring system shall be considered within the limits of validation when the maximum air flow reading during testing does not exceed the highest validated air flow value by 20 %. The air flow measuring system shall be considered to be all piping and test chamber elements from the air flow measuring device to the orifice plate.

9.6 The measured flow at each listed pressure for each orifice plate shall be determined with an error not greater than ± 5 % when the flow is greater than 0.944 L/s [2 ft³/min] or ± 10 % when the flow is less than 0.944 L/s [2 ft³/min] but greater than 0.236 L/s [0.5 ft³/min].

NOTE 5—At lower flows, a greater percentage of error will usually be acceptable. If higher precision is required, special flow measuring techniques are necessary. The accuracy of the specimen leakage flow measurement is affected by the accuracy of the flow meter and the amount of extraneous leakage of the apparatus (see Annex A1).

9.7 Alternate means may be used for validating the air flow measuring system as long as they can be proven to provide the same level of accuracy and are traceable to NIST.

TABLE 1 Orifice Plates Data

Orifice Plate Hole Sizes	Nominal Flow	Differential Pressure Across Orifice Plate
6.35 mm [0.25 in.]	0.22 L/s [0.47 ft ³ /min]	75 Pa [1.57 psf]
12.7 mm [0.5 in.]	0.88 L/s [1.86 ft ³ /min]	75 Pa [1.57 psf]
25.4 mm [1.0 in.]	3.47 L/s [7.36 ft ³ /min]	75 Pa [1.57 psf]
38.1 mm [1.5 in.]	7.66 L/s [16.24 ft ³ /min]	75 Pa [1.57 psf]
50.8 mm [2.0 in.]	13.64 L/s [28.90 ft ³ /min]	75 Pa [1.57 psf]
63.5 mm [2.5 in.]	21.33 L/s [45.20 ft ³ /min]	75 Pa [1.57 psf]

9.8 Validation shall be performed at least once every six months using the method described above. Alternative orifice mounting conditions may be used during interim validation periods for air flow checking purposes, but these methods shall be documented in the laboratory's Operation Manual.

10. Test Conditions

10.1 The specifying authority shall supply the following information:

10.1.1 Specimen test size,

10.1.2 Test pressure difference (if no value is designated, 75 Pa [1.57 lb/ft²]),

10.1.3 Direction of air flow, exfiltration or infiltration. (If none is specified, the test shall be infiltration.), and

10.1.4 Units of air leakage rate by operable crack perimeter or unit area. If none is specified, report both.

11. Procedure

11.1 Eliminate extraneous chamber leakage. If this is impractical, measure the amount of leakage with the specimen sealed at the air pressure differences to be exerted during the air leakage tests. Designate this measured air flow as the extraneous air flow, Q_e .

11.2 Remove any sealing material or construction that is not normally a part of the specimen assembly as installed in or on a building. Fit the specimen into or against the chamber opening. Installation should be such that no parts or openings of the specimen are obstructed. The specimen shall be installed in accordance with the manufacturer's documented instructions and, in the case of skylights, at the lowest slope recommended by those instructions. In absence of documented instructions, skylight products shall be tested in the horizontal position.

NOTE 6—Nonhardening mastic compounds or pressure sensitive tape can be used effectively to seal the test specimen to the chamber, and to achieve air tightness in the construction of the chamber. These materials can also be used to seal a separate mounting panel to the chamber. Rubber gaskets with clamping devices may also be used for this purpose, provided that the gasket is highly flexible and has a narrow contact edge.

11.3 Without disturbing the seal between the specimen and the test chamber, adjust all hardware, ventilators, balances, sash, doors, and other components included as an integral part of the specimen so that their operation conforms to test method requirements.

11.4 To ensure proper alignment and weather seal compression, fully open, close, and lock each ventilator, sash, or door five times prior to testing.

11.5 Adjust the air flow through the test chamber to provide the specified test pressure difference across the test specimen. When the test conditions are stabilized, record the air flow through the flow meter and the test pressure difference. This measured air flow is designated the total air flow, Q_t . The recorded measurements shall include the absolute pressure, the temperature, and the relative humidity of the air at the locations described in 6.2.

11.6 If necessary, use the measurements of the temperature, relative humidity, and the absolute pressure to adjust the

measured flow rates into flow rates at standard atmospheric conditions using the calculation procedures outlined in Section 12.

12. Calculation

12.1 Air flow measurements shall be reported at standard conditions. There are two different means of measuring air flow, volumetric flow measurements, and mass flow measurements. Depending on the instrument used, measurements from some volumetric flow meters may need to be adjusted to report flow rates at standard conditions. Both the viscosity and the density of air vary with the pressure, temperature, and relative humidity (RH) of the air flowing through the flow meter, and 12.2 describes how to adjust those measurements to standard conditions. Some manufactures of Laminar Flow Elements (flow meters) provide means of adjusting the results based on the actual conditions in the laboratory, and in those circumstances the instrument manufacturers' methodology shall be used. For mass flow measurements, the user shall use the methodology provided by the manufacturer to adjust the air flow rate to standard conditions. If the instrument manufacturer does not provide means of adjusting the results to standard environmental conditions, use the following equations to make this adjustment.

12.2 *Converting Volumetric Flow Rate Measurements to Standard Conditions*—Actual volumetric flow rate measurements are adjusted to standard volumetric flow rates using:

$$Q_t = Q \cdot (\mu_{std} / \mu_t) \cdot (T_{std} / T_t) \cdot (P_t / P_{std}) \cdot (\rho_{wet} / \rho_{dry}) \quad (1)$$

where:

- Q_t = air leakage through the test specimen at standard conditions, L/s [ft³/min],
- Q = measured air leakage through the test specimen at actual conditions, L/s [ft³/min],
- μ_{std} = viscosity at standard conditions, $\mu\text{Pa}\cdot\text{s}$,
- μ_t = viscosity at actual conditions, $\mu\text{Pa}\cdot\text{s}$,
- T_{std} = temperature at standard conditions, K (293.95 K),
- T_t = measured temperature at actual conditions, K,
- P_t = measured barometric pressure at actual conditions, kPa,
- P_{std} = barometric pressure at standard conditions, kPa (101.3 kPa),
- ρ_{wet} = density of air at actual conditions, kg/m³, and
- ρ_{dry} = density of dry air, kg/m³ (1.202 kg/m³).

12.2.1 *Viscosity Correction*—The following equation shall be used to correct for viscosity changes due to temperature:

$$(\mu_{std} / \mu_t) = ((T_t + 110.4) / 413.95) \cdot (T_t / 293.95)^{2.3} \quad (2)$$

12.2.1.1 The change to the viscosity of air due to humidity is less than 1.5 % at temperatures below 40 °C and at relative humidity levels below 90 %. Corrections to viscosity for relative humidity are not required unless the temperature and relative humidity are above those values. See the manufacturers' recommendations for correcting for viscosity of wet air that is outside of those ranges.

12.2.2 *Temperature Correction*—The temperature correction to account for the change in air volume can be predicted by the following equation:

$$(T_{std} / T_t) = 293.95 / (T_t + 273.15) \quad (3)$$

12.2.3 *Pressure Correction*—The pressure correction to account for the change in density with pressure is predicted by:

$$(P_t / P_{std}) = (P_t / 101.325) \quad (4)$$

12.2.4 *Humidity Correction*—The correction to adjust for the density of air due to the variation in the relative humidity of air is found in Eq 5 and Eq 6:

$$(\rho_{wet} / \rho_{dry}) = (P_t - P_y) / (R_d \cdot T_t) + (P_y / R_v \cdot T_t) / (P_t / R \cdot T_t) \quad (5)$$

where:

$$P_y = RH \cdot 6.1078 \cdot 10^{\wedge}(7.5 \cdot (T_t - 275.15) / (T_t - 35.85)) \quad (6)$$

where:

- ρ_{wet} = density of wet air, kg/m³,
- ρ_{dry} = density of dry air, kg/m³,
- P_t = barometric pressure, Pa,
- RH = relative humidity, RH%, (that is, 50.0),
- R_d = specific gas constant of dry air, J/(kg·K), (287.058 J/(kg·K)),
- R_v = specific gas constant of water vapor, J/(kg·K), (461.495 J/(kg·K)), and
- T_t = measured temperature of air, K.

12.3 *Extraneous Air Flow*—The actual air leakage through the test specimen, Q_s , is found by subtracting the extraneous air flow, Q_e , from the measured total air flow, Q_t , as follows:

$$Q_s = Q_t - Q_e \quad (7)$$

12.4 *Air Leakage Rate*—Calculate the rate of air leakage for the test specimen according to 12.4.1 and 12.4.2.

12.4.1 To calculate the rate of air leakage per unit of length of operable crack perimeter, q_l , use Eq 8:

$$q_l = Q_s / l \quad (8)$$

where:

- l = operable crack perimeter, m [ft],
- Q_s = airflow corrected to standard conditions, L/s [ft³/min], and
- q_l = rate of air leakage per unit length of operable crack perimeter, L/(s·m) [ft³/min·ft].

12.4.2 To calculate rate of air leakage per unit area, q_A :

$$q_A = Q_s / A \quad (9)$$

where:

- A = area, m² [ft²],
- Q_s = airflow corrected to standard conditions, L/s [ft³/min], and
- q_A = rate of air leakage per unit area, L/(s·m²) [ft³/min·ft²].

13. Report

13.1 Report the following information:

13.1.1 *General*—Testing agency, date and time of test, and date of report.

13.1.2 *Sample Description*—Manufacturer, model, operation type, materials, and other pertinent information; description of the locking and operating mechanisms if applicable; glass thickness, type and method of glazing; weather seal dimensions, type, weep holes, and material; and crack perimeter or specimen area, or both.

13.1.3 Drawings of Specimen—Detailed drawings of the specimen showing dimensioned section profiles, sash or door dimensions and arrangement, framing location, panel arrangement, installation and spacing of anchorage, weatherstripping, locking arrangement, hardware, sealants, glazing details, angle from the horizontal for skylights, weep holes, drainage path detail, and any other pertinent construction details. Any modifications made on the specimen to obtain the reported test values shall be noted.

13.1.4 Location of Air Seal—Detailed drawing showing the air seal between the test specimen and the test chamber or mounting frame. The drawing shall clearly indicate the location of the air seal relative to the specimen frame.

13.1.5 Test Parameters—List or describe the specified test pressure difference(s), whether the tests were conducted for infiltration or exfiltration, and whether a positive or negative test pressure was used. List air flow measurement recorded conditions (that is, absolute pressure, temperature, and relative humidity (RH)).

13.1.6 Pressure Differences—A statement or tabulation of the pressure differentials exerted across the specimen during the test.

13.1.7 Air Leakage Rate—Air leakage rate shall be reported per unit length of operable crack perimeter (q_l) or per unit area (q_A), or both. Specimen air leakage rate (Q_s).

13.1.8 Compliance Statement—A statement that the tests were conducted in accordance with this test method, or a complete description of any deviation from this test method. When the tests are conducted to check for conformity of the specimen to a particular performance specification, the specification shall be identified.

13.2 If several identical specimens are tested, the results for each specimen shall be reported, each specimen being properly identified, particularly with respect to distinguishing features or differing adjustment. A separate drawing for each specimen shall not be required if all differences between the specimens are noted on the drawings provided.

14. Precision and Bias

14.1 The precision and bias of this test method has not been determined.

15. Keywords

15.1 air leakage; curtain walls; doors; fenestration; laboratory method; roof windows; skylights; sloped glazing system; static pressure chamber; TDD; tubular daylighting devices; windows

ANNEX

(Mandatory Information)

A1. ERRORS IN WINDOW AIR LEAKAGE MEASUREMENT

A1.1 Terminology

A1.1.1 Symbols:

A1.1.1.1 Q_s = air flow through specimen

A1.1.1.2 Q_t = total air flow

A1.1.1.3 Q_e = extraneous air flow

A1.1.1.4 Δ = delta

NOTE A1.1—Symbols **A1.1.1.1 – A1.1.1.4** have been converted to standard conditions.

A1.2 In the apparatus using a supply air system, $Q_s = Q_t - Q_e$, the extraneous air leakage (Q_e) represents all the air leakage leaving the chamber which does not pass through the specimen proper. This includes leakage passing through the chamber walls and around the specimen mounting. When the mounting panel is used, leakage between the chamber and the panel contributes to extraneous leakage. The extraneous leakage flow is a function of the pressure difference between the chamber and the room, which is also the test specimen difference.

A1.3 The total error in the specimen flow determination (neglecting errors in the air density determination) is as follows:

$$\Delta Q_s / Q_s = [\Delta Q_t / (Q_t \cdot Q_e)] \pm [\Delta Q_e / (Q_t \cdot Q_e)] \quad (\text{A1.1})$$

A1.3.1 According to **6.2.3**, the air flow through the test specimen is to be determined with an error no greater than $\Delta Q_s / Q_s = \pm 5\%$ if the extraneous leakage is accurate to $\Delta Q_e / Q_e = \pm 10\%$ and Q_e is 10 % of Q_s , then the contribution of the extraneous leakage to the overall error in **Eq 2** is $\pm 1\%$. (Note that the error attributed to the extraneous leakage determination is a function not only of the accuracy of the flow meter used in the determination, but also of the constancy of the leakage from the time of determination to the time of test.) The error contributed by the flow meter to the total error is then limited to 4 %, but because $Q_t = Q_s + Q_e = 1.10 Q_s$ the accuracy required of the flow meter is:

$$\Delta(Q_s / Q_t) = 4/1.1\% = 3.6\% \quad (\text{A1.2})$$

A1.3.2 It is seen that the major factor affecting the accuracy required of the flow meter is the proportion of Q_e to Q_s . If $\Delta Q_e / Q_e$ remains at $\pm 10\%$, but Q_t is 150 % of Q_s , the error contributed by the extraneous leakage becomes 5 % and no error can be tolerated in the flow meter if the conditions of **6.2.3** are to be met—with Q_e in excess of 50 % of Q_s , it is impossible to achieve the required overall limit of error. Likewise, if the extraneous leakage is eliminated, the flow meter error can be as great as 5 %.



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